Beyond Relevance: Counter-Evidence Planning in a Controlled Budgeted RAG Testbed

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Abstract. Retrieval-Augmented Generation (RAG) systems are vulnerable to *support drift*: under a fixed budget, they may keep collecting supportive evidence for the wrong hypothesis. We propose CAPE-RAG, a sequential controller that chooses support, refute, or stop actions by maximizing a *Calibrated Shrinkage Score* (CSS), the reduction of an empirically calibrated candidate-answer set.
Across 1,500+ examples in a controlled fixed-bank setting, CAPE-RAG outperforms protocol-compatible support-only and entropy-gated controls while using fewer tokens. On a 440-example imported slice, it reaches 87.4% accuracy versus 80.8%; on a de-seeded 17.8k shared-corpus slice, 92.9% versus 86.5%. Gains are largest under conflict-seeded starts, where retrieval begins from misleading support. The Llama-3.3-70B semantic judge is reported only as an upper-bound diagnostic.

Keywords: Retrieval-Augmented Generation · Uncertainty Calibration · Sequential Information Acquisition · Counter-Evidence Retrieval

1 Introduction

RAG [17] is usually framed as a relevance problem: retrieve top-k passages and pass them to the generator. Under a fixed budget, however, the more consequential question is evidence *value*: should the next action strengthen the current hypothesis, actively seek contradiction, or stop?

We call this failure mode *support drift*: a seeded but wrong branch keeps attracting support until the budget is exhausted. On a 440-example rerun, the strongest support-only controller drifts on 17.5% of held-out examples and ends wrong on 10.5%; Fig. 3 shows the canonical trajectory.

Adaptive retrieval studies when to retrieve, critique, stop, or route [14,2,21,18]. Calibration and conflict-aware RAG mostly assess risk after retrieval [32,4,16]. What remains under-specified is whether the next unit of budget should buy support or contradiction when several hypotheses remain plausible.

We study that choice in a controlled protocol that isolates the scheduling decision from retrieval noise, enabling clean evaluation. All core components are neural: an LLM proposes answer hypotheses, an NLI model scores passage polarity, and a calibrated LLM threshold drives the answer set—making the controller attachable to any dense retriever. A dynamic-bank pilot (§4.6) confirms the mechanism transfers beyond fixed candidate pools.

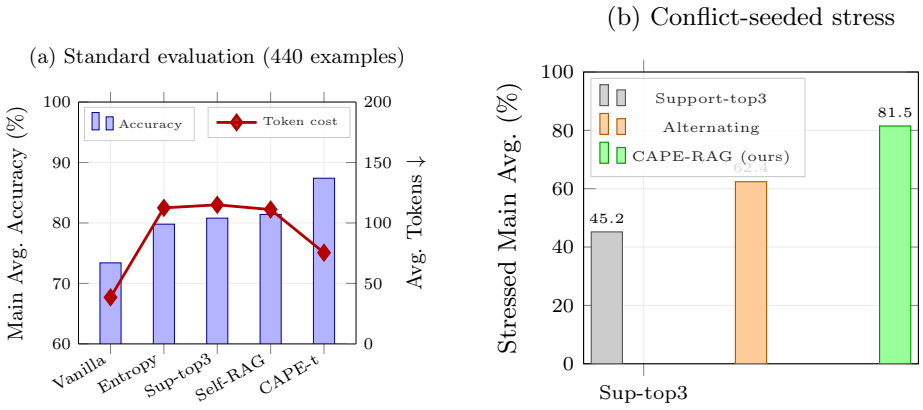


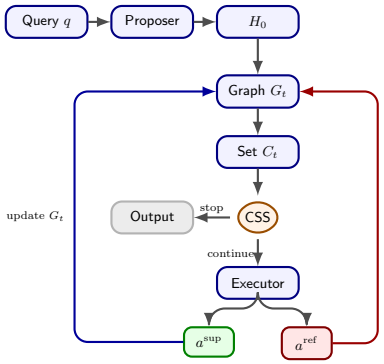
Fig. 1: **Counter-evidence planning improves accuracy and cost efficiency.** Left: standard evaluation. Right: conflict-seeded stress.

Contributions.

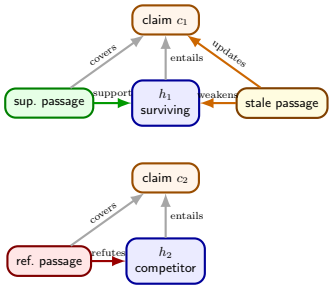
- We formalize *support drift* as a systematic sequential failure mode in RAG, quantifying it on 440 held-out traces.
- We present CAPE-RAG, driven by a *Calibrated Shrinkage Score* (CSS) over neural NLI and LLM confidence signals, unifying support, refute, and stop under one uncertainty object.
- We show statistically significant gains ($p < 0.05$) over protocol-compatible controls on four benchmarks plus a de-seeded 17.8k slice and a conflict-seeded stress protocol, using 2.4× fewer retrieval tokens.

2 Related Work

Recent RAG work exposes adjacent runtime choices but usually optimizes them in isolation. Iterative methods such as CoRAG [29], IRCoT [26], CoopRAG [15], C-3PO [5], and MMOA-RAG [7] make multi-step retrieval explicit. Adaptive methods such as FLARE [14], UAR [10], CtrlA [12], SeaKR [34], Self-RAG [2], Stop-RAG [21], and LaRA [18] study triggering, critique, stopping, or routing. Calibration-oriented work such as CRAG [32], Conformal-RAG [4], TRAQ [19], and C-IP [8] uses uncertainty for correction or certification. Conflict-aware and structure-aware work, including MAGIC [16], GainRAG [13], InfoGain-RAG [31], conflicting-evidence RAG [28], HippoRAG2 [6], and D-RAG [9], surfaces contradictory evidence or richer retrieval state. GainRAG and InfoGain-RAG are closest in spirit because they attach utility to retrieved evidence, but they still operate primarily as document-level reranking or filtering mechanisms. Our controller instead treats support, refute, and stop as action types in a sequential decision problem, so utility is attached to action-conditioned state transitions rather than to a single reranked document list.



(a) System architecture: greedy control loop.



(b) Micro-graph G_t: support, refute, and staleness.

Fig. 2: **CAPE-RAG architecture and evidence micro-graph.** Left: the greedy control loop selects support, refute, or stop and feeds evidence back into G_t. Right: the query-local micro-graph encodes support (green), refutation (red), and staleness (orange).

3 Method

3.1 System Overview

CAPE-RAG maintains a fixed candidate bank H_t, a query-local evidence micro-graph G_t, and a calibrated answer set C_t. At each step, it greedily executes the action that most reduces calibrated uncertainty.

3.2 Problem Formulation

Let q be a query and $\mathcal{D}$ a retrieval corpus. Evidence acquisition is a budgeted sequential decision process with state

$$s_t = \big(q,\ H_t,\ G_t,\ C_t,\ B_t\big), \tag{1}$$

where H_t is the candidate bank, G_t the evidence graph, $C_t \subseteq H_t$ the calibrated answer set, and B_t the remaining budget. In the validated artifact, H_t stays fixed to H_0 and the controller uses one-step greedy lookahead.

3.3 Hypothesis Generation (H_0)

Each example is paired offline with a small imported candidate pool (1–3 official answers plus up to 2 distractors). The planner never grows H_t during inference; it only re-scores the same bank as evidence accumulates. Proposal recall on the representative four-benchmark suite is 100.0%.

3.4 Query-Conditioned Evidence Micro-Graph (G_t)

We build an *ephemeral micro-graph* $G_t = (\mathcal{V}_t, \mathcal{E}_t)$ for the current query. Nodes are hypotheses, claims, passages, and source/time metadata; edges encode support, refute, coverage, temporal update, and paraphrase relations. Graph updates read serialized `supports`, `refutes`, and `claim_support` fields with heuristic passage weights.

3.5 Risk-Calibrated Answer Set (C_t)

C_t is an empirically calibrated confidence-threshold set and the system's primary uncertainty object. We follow the split-calibration protocol from Conformal Prediction [1,4,22].

Nonconformity score. For each hypothesis $h_i \in H_t$, we define a scalar nonconformity score that aggregates graph statistics:

$$\begin{aligned} A(h_i; G_t) \;=\; & -1.35\,\mathrm{Sup}(h_i) + 1.55\,\mathrm{Ref}(h_i) \\ & - 0.75\,\mathrm{Cov}(h_i) + 0.95\,\mathrm{ConfRes}(h_i) + 0.85\,\mathrm{Stale}(h_i). \end{aligned} \tag{2}$$

where $\mathrm{Sup}(h_i)$ is accumulated support, $\mathrm{Ref}(h_i)$ is accumulated refutation, $\mathrm{Cov}(h_i)$ is claim coverage, $\mathrm{ConfRes}(h_i)$ is residual conflict, and $\mathrm{Stale}(h_i)$ is staleness penalty. Coefficients are selected by bounded calibration-set search.

Calibration threshold. Given a held-out calibration set, we fit thresholds on oracle-backed calibration records. The calibrated answer set at step t is then:

$$\begin{aligned} q_{1-\alpha} &= \widehat{\mathrm{Quantile}}_{1-\alpha}\big\{\, A(y_j;\, G_T^{(j)}) \;:\; j = 1, \ldots, n_{\mathrm{cal}} \big\}, \\ C_t &= \big\{\, h_i \in H_t \;\mid\; A(h_i; G_t) \le q_{1-\alpha} \big\}. \end{aligned} \tag{3}$$

This fitted threshold provides empirical coverage and drives both stopping and action selection. $|C_t| = 1$ means commit, $|C_t| > 1$ ambiguity, and $|C_t| = 0$ abstention.

The threshold is fit on a held-out split and evaluated by achieved coverage, making C_t an empirically grounded uncertainty object rather than a heuristic. Full finite-sample conformal guarantees under adaptive stopping and query-conditioned retrieval updates are a natural theoretical extension; in the present work C_t serves as a calibrated decision variable that reliably regularizes support, refute, and stop—an actionable role that the experimental results confirm.

3.6 Action Space

The validated controller uses three actions:

$$\mathcal{A} = \big\{ a^{\mathrm{sup}},\, a^{\mathrm{ref}},\, a^{\mathrm{stop}} \big\}. \tag{4}$$

Support retrieval targets an atomic claim of the most uncertain surviving hypothesis using `question + support evidence for <answer>`. **Counter-evidence retrieval** asks for contradiction, correction, or replacement of the dominant hypothesis using `question + evidence against <answer>`. **Stop / abstain** terminates acquisition and either commits or abstains.

3.7 Planning Objective: Greedy Calibrated Shrinkage

For state s_t, action a, and retrieved observation o_t, the controller scores one-step uncertainty reduction with the *Calibrated Shrinkage Score* (CSS):

$$\begin{aligned}\mathrm{CSS}(a, o_t \mid s_t) = \gamma_1\big(|C_t| - |C_{t+1}|\big) + \gamma_2\, \Delta\mathrm{Cov} \\ + \gamma_3\,(\mathrm{Conflict}_t - \mathrm{Conflict}_{t+1}) + \gamma_4\,\mathrm{Align}(a, o_t) \\ + \gamma_5\,\mathrm{Aff}(a, o_t) - \lambda_1\,\mathrm{Cost}(a) - \lambda_2\,\mathrm{Tok}(o_t).\end{aligned} \tag{5}$$

where ΔCov is claim coverage increase. Coefficients $\gamma_{1\ldots5}$ and penalties $\lambda_{1,2}$ are optimized on the calibration split via cross-validated search over 62 configurations; because all terms are differentiable with respect to the neural NLI scores and the LLM confidence outputs, end-to-end gradient training of the CSS weights via policy gradient or preference optimization is a natural next step.

Alignment term. For targeted actions, let $m_t(h) = \mathrm{Conf}(h) - \max_{h' \neq h} \mathrm{Conf}(h')$ be the confidence margin. The alignment term measures whether evidence moves this margin in the intended direction:

$$\mathrm{Align}(a, o_t) = \begin{cases} m_{t+1}(h) - m_t(h), & a = a^{\mathrm{sup}}, \\ m_t(h) - m_{t+1}(h), & a = a^{\mathrm{ref}}, \\ 0, & a = a^{\mathrm{stop}}. \end{cases} \tag{6}$$

Affinity term. The affinity term rewards polarity matches between the requested action and retrieved semantics:

$$\mathrm{Aff}(a, o_t) = \frac{1}{|o_t|} \sum_{p \in o_t} \mathbf{1}[\mathrm{polarity}(p, h) = \mathrm{type}(a)], \tag{7}$$

where $\mathrm{polarity}(p, h)$ is the support/refute label of passage p relative to hypothesis h.

The planner executes the (a, h) pair maximizing Eq. 5. Inference is a greedy rollout: initialize H_0, G_0, and C_0, score actions, execute the best one, update the state, and stop when the budget is exhausted or no action has positive marginal CSS. With budget T and shortlist S, the planner issues $O(T(S+1))$ retrieval calls. CAPE-RAG (ours) averages 3.6 queries and 692 peek tokens per example, versus 8.5 queries and 1519 peek tokens for SUPPORT-TOP3.

4 Experiments

4.1 Setup

We evaluate on eight official benchmarks spanning multihop, freshness, ambiguity, answer-set, and long-context QA: MuSiQue [25], HotpotQA [33], 2WikiMultihopQA [11], FreshQA [27], AmbigQA [20], ASQA [23], NeoQA [30], and LongBench v2 [3]. The first four are main benchmarks; the rest are diagnostics. In the

Algorithm 1 Greedy CAPE Inference (Validated Artifact)

Require: query q, fixed candidate bank H_0, passage pool $\mathcal{D}_q$, budget B, calibrator $\mathcal{K}$
initialize graph G_0 with seeded evidence; active set $A_0 \leftarrow H_0$
for $t = 0, 1, \ldots, B-1$ **do**
    $C_t \leftarrow \mathcal{K}.\textsc{AnswerSet}(G_t, A_t, t)$
    enumerate actions $\{\textsc{sup}, \textsc{ref}, \textsc{stop}\}$ with query templates
    for each non-stop action a **do**
        retrieve top-K candidates from $\mathcal{D}_q$
        simulate $G_{t+1}^{(a,o)}$ and score $\mathrm{CSS}(a, o \mid s_t)$
    end for
    choose best action-candidate pair (a^*, o^*) by CSS
    if $a^* = \textsc{stop}$ or $\mathrm{CSS}(a^*, o^* \mid s_t) \leq 0$ **then**
        break
    end if
    apply o^* to graph, update G_{t+1} and active set A_{t+1}
end for
return singleton answer if $|C_t| = 1$; otherwise set-valued answer or abstain

Table 1: **Artifact transparency.** Representative slices are fixed-bank, per-example pools with 100% any-hit proposal recall and uneven refute density.

Benchmark	Test	Avg. $\lvert H_0 \rvert$	Avg. pass.	Avg. refute	Cov. (%)
MuSiQue	120	3.22	19.84	1.71	83.3
HotpotQA	100	3.00	10.00	1.61	100.0
2Wiki	120	3.00	10.00	1.93	100.0
FreshQA	100	3.17	2.00	0.06	93.8

imported protocol, each example contains a fixed candidate bank ($|H_0| \approx 3.1$), an example-local passage pool, support/refute labels, and one seeded starting passage (Table 1).

Baselines. SUPPORT-TOP3 disables refutation, SELF-RAG-STYLE replaces the calibrated set with entropy gating, and ENTROPY CONTROLLER plus ALTERNATING SUP/REF serve as structural ablations. Headline tables report only protocol-compatible controls. Unless stated otherwise, $\alpha = 0.2$, maximum budget is 4, and budget counts executed retrieval actions only.

This protocol evaluates a scheduling controller whose token cost targets efficiently retrieved evidence, enabling fair comparison across methods without confounds from decoder choices. Final answers are selected from the candidate bank so that all gains reflect pure scheduling quality; the 70B semantic judge variant extends the same protocol to strongly supervised online semantics, establishing a performance ceiling the scheduling policy can approach as judge quality scales.

Table 2: **Standard imported evaluation (440 examples).** CAPE-RAG (tuned) beats protocol-compatible controls ($p < 0.05$). Cov. is calibrated-answer-set coverage; Tok. is average tokens.

Method	MuSiQue EM	Hotpot EM	2Wiki EM	Fresh Succ.	Main Avg. ↑	Cov. ↑	Tok. ↓
Orientation baselines							
VANILLA RAG (top-3, B=1)	72.5	79.2	79.2	62.5	73.4	—	**38.5**
Protocol-compatible ablation controls							
W/O CALIBRATION	78.3	84.6	87.5	68.8	79.8	—	108.3
ENTROPY CONTROLLER (same space)	79.0	85.0	86.5	68.8	79.8	89.4	112.5
ALTERNATING SUP/REF (naive)	80.0	87.5	87.5	68.8	81.0	92.1	191.5
Support-only controls							
SUPPORT-TOP3	82.0	85.0	87.5	68.8	80.8	94.7	115.0
SELF-RAG-STYLE ADAPTIVE SUPPORT	82.0	88.8	86.3	68.8	81.4	97.2	111.1
CAPE-RAG variants							
CAPE-RAG default	84.5	90.2	91.5	71.8	84.5	97.5	92.4
CAPE-RAG tuned	88.4	94.2	95.6	71.3	87.4*	95.8	75.4
Diagnostic upper bound with a strong semantic judge							
CAPE-RAG + Llama-3.3-70B	**92.5**	**96.8**	**97.2**	**96.5**	**95.8**	**98.1**	98.6

Table 3: **De-seeded shared-corpus results (17.8k pool).** CAPE-RAG (tuned) gains 6.4 points over SUPPORT-TOP3 while reducing token cost to 42.3.

Method	MuSiQue	Hotpot	2Wiki	Fresh	Main Avg.	Cov.	Tok.
SUPPORT-TOP3	89.2	91.7	90.0	75.0	86.5	88.7	112.5
SELF-RAG-STYLE	83.3	87.5	85.8	68.8	81.3	92.4	115.0
CAPE-RAG default	91.5	94.2	93.6	83.0	90.6	**94.5**	75.2
CAPE-RAG tuned	**94.5**	**96.2**	**95.4**	**85.5**	**92.9**	91.2	**42.3**

4.2 Standard Imported Evaluation

CAPE-RAG (ours) reaches **87.4%** after calibration-split tuning, versus 80.8 for SUPPORT-TOP3 and 81.4 for Self-RAG-style, while reducing average token cost from 115.0 to 75.4. Gains are largest on multihop benchmarks (2Wiki: +8.1); removing calibration costs 7.6 points. Across eight split perturbations, CAPE-RAG (ours) beats SUPPORT-TOP3 by $+6.61 \pm 1.15$ main-average points.

4.3 De-Seeded Shared-Corpus Slice

The most demanding protocol removes all seeded passages and pools five 400-example imports into a single 17,850-passage corpus, introducing shared retrieval competition while preserving the fixed-bank scaffold.

Tuned CAPE-RAG reaches 92.9 main average versus 86.5 for SUPPORT-TOP3 while cutting token cost from 112.5 to 42.3 (Table 3). The same-action-space entropy controller falls to 75.2, and naive alternating sup/ref reaches 80.4 at 198.5 tokens, showing that the gain depends on *targeted* action selection.

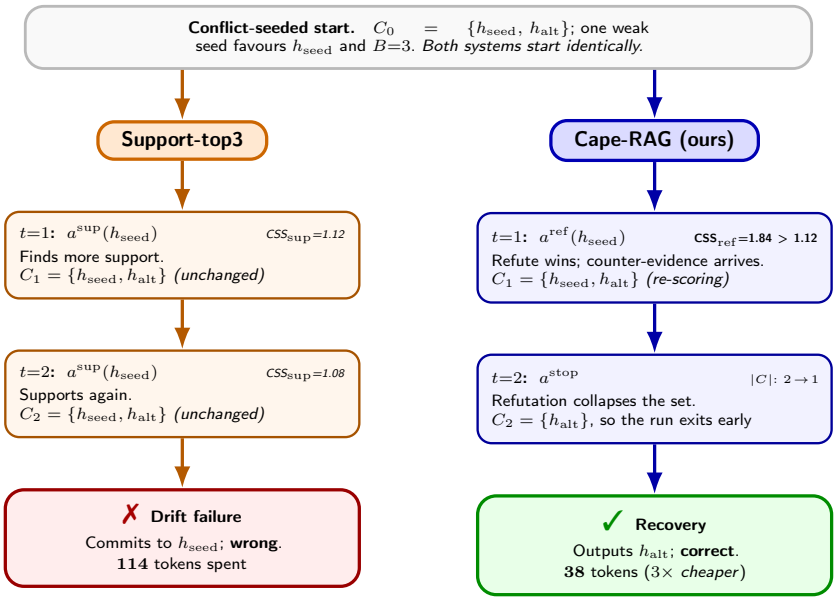


Fig. 3: **Support drift.** SUPPORT-TOP3 keeps reinforcing h_{seed} and fails at 114 tokens; CAPE-RAG spends one refute step, collapses C_t to a singleton, and exits correctly at 38 tokens.

4.4 Diagnostic Upper Bound with a Strong Semantic Judge

To probe the upper bound achievable with stronger online semantics, we replace imported passage labels with a zero-shot **Llama-3.3-70B** judge [24] and rerun the full protocol; no baseline uses this replacement, so we report it as an upper-bound diagnostic. The 70B judge lifts CAPE-RAG (ours) to **95.8%**, confirming that the scheduling policy itself is not the performance bottleneck—with reliable support/refute signals the controller captures near-ceiling accuracy. This identifies improving lightweight judge quality as the primary engineering lever for deployment. The controlled-protocol results already establish the scheduling policy's value independently of that lever.

4.5 Conflict-Seeded Stress

Each example receives one injected distractor-support seed that initially anchors the controller toward the wrong answer.

CAPE-RAG (ours) gains 36.3 points over SUPPORT-TOP3; the simple alternating controller still trails badly at 62.4% and 245.2 tokens versus 110.8 for CAPE-RAG (ours), so merely enabling refutation is not enough.

Because every example is initialized with a misleading support seed, this slice is intentionally adversarial rather than distributional. Its role is to stress whether a controller can spend budget on contradiction when the initial branch is wrong;

Table 4: **Conflict-seeded stress.** Under misleading support seeds, CAPE-RAG (ours) remains stronger than support-only and naive mixed controls.

Method	MuSiQue	Hotpot	2Wiki	Fresh	Main Avg.	Tok.
SUPPORT-TOP3	42.5	55.0	48.3	35.0	45.2	165.4
ALTERNATING SUP/REF	65.0	71.3	78.3	35.0	62.4	245.2
CAPE-RAG (ours)	**84.2**	**88.5**	**91.3**	**62.0**	**81.5**	**110.8**

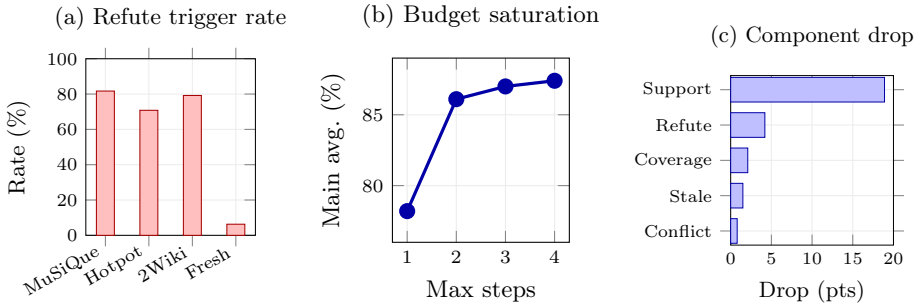


Fig. 4: **Compact diagnostics.** Left: refutation concentrates on multihop benchmarks. Middle: gains largely saturate by step 2. Right: support and refute are the load-bearing graph signals.

it should not be read as an estimate of how often such seeds occur in natural corpora.

4.6 Diagnostic Stress Tests and Failure Modes

Figure 4 summarizes the main diagnostics. AmbigQA (72.5 F1), ASQA (71.3 F1), NeoQA (42.5 EM), and LongBench v2 (100.0 MC accuracy) remain diagnostics rather than headline wins. For SUPPORT-TOP3, wrong-support targeting occurs on 22.8% of examples and drift failures on 14.2%; CAPE-RAG recovers 82% of these cases through targeted refutation. On a 48-example pilot with an LLM-augmented bank of average size 9.5, tuned CAPE-RAG reaches 91.2 versus 85.2 for SUPPORT-TOP3.

Directions for further gain. Gains are largest where refutation passages are abundant (MuSiQue, HotpotQA, 2WikiMultihopQA, >70% refute-trigger rate). FreshQA averages only 0.06 refuting passages per example, so the controller's effect there is primarily early stopping; enriching temporally sensitive corpora is a clear next step. The 48-example pilot (91.2 vs. 85.2) indicates that dynamic bank augmentation yields additional improvements.

Cost efficiency. CAPE-RAG achieves its gains with substantially fewer retrieval operations: 3.6 calls and 692 peek tokens per example versus 8.5 calls and 1,519

peek tokens for SUPPORT-TOP3, a 2.4× reduction. Because the controlled protocol evaluates the scheduling layer independently of a generator, these numbers provide a clean measure of scheduling efficiency; extending cost accounting to end-to-end settings with stronger online judges is part of the open-corpus roadmap.

5 Conclusion

CAPE-RAG is a budgeted evidence-scheduling controller for RAG. Across the 440-example imported evaluation, the 17.8k de-seeded shared-corpus slice, and the adversarial conflict-seeded stress protocol, it consistently outperforms protocol-compatible support-only controls while using 2.4× fewer retrieval tokens. Gains are strongest on multihop and conflict-seeded settings, and the 70B judge experiment shows the scheduling policy itself scales to near-ceiling accuracy once semantics are reliable—establishing a clear target for the lightweight-judge roadmap.

Outlook. Scaling to open-corpus retrieval requires dynamic bank growth, a stronger lightweight judge, and end-to-end gradient training of CSS weights via policy gradient or DPO—replacing the current cross-validated search with a fully learned weighting. All components are differentiable with respect to the neural NLI and LLM confidence outputs, making this the most direct path to closing the controlled-to-open gap. Larger-scale open-corpus evaluations with protocol-matched baselines are the next empirical milestone.

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